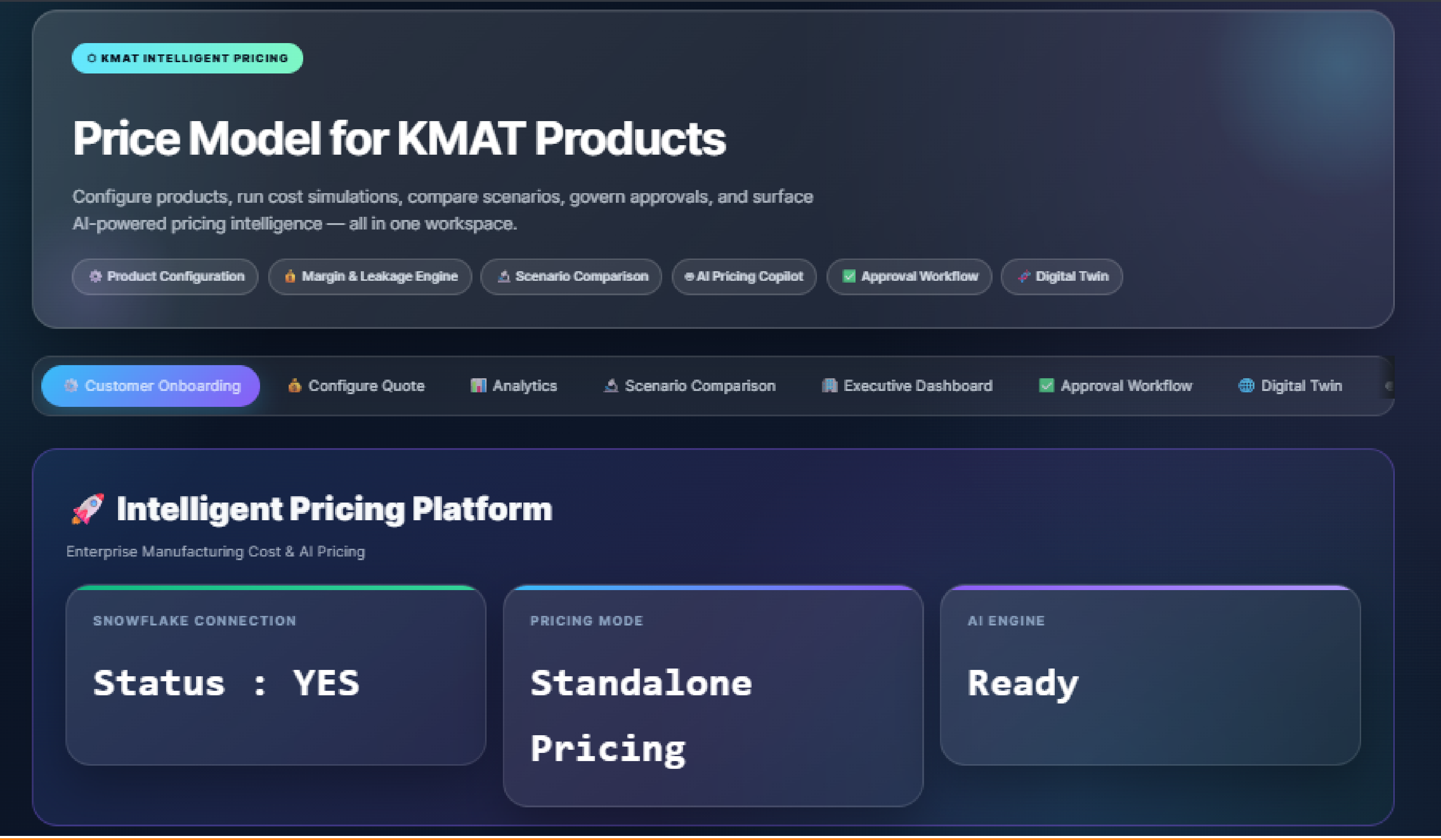

Price Model for KMAT Products

Project Documentation

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Date: July 20, 2026

An AI-powered platform to calculate manufacturing costs, generate optimized prices, and prevent revenue leakage.

Contents

Chapter 1 Project Overview

Introduction

- *What the Intelligent Pricing Platform is*
- *The core objectives of the project*
- *Who it is built for*

1.1 What is the Intelligent Pricing Platform

The Intelligent Pricing Platform is a web-based application that helps manufacturing and product-based businesses calculate accurate manufacturing costs, generate optimized selling prices, and protect profit margins.

The platform brings together cost data, pricing rules, and AI-based recommendations in a single dashboard, so that pricing teams no longer have to rely on scattered spreadsheets.

1.2 Project Objectives

- Automate manufacturing cost calculation for products and product families.
- Recommend optimized selling prices based on cost, margin targets, and market factors.
- Allow side-by-side comparison of different pricing scenarios.
- Detect revenue leakage caused by discounts, errors, or approval gaps.
- Provide a single executive dashboard for pricing visibility.
- Support a formal approval workflow for price changes.
- Provide a digital twin view to simulate the impact of cost or price changes before they go live.
- Provide an AI pricing copilot that answers pricing questions in plain language.

1.3 Intended Users

Table 1.1: Primary Users of the Platform

User Role	What They Use the Platform For
Pricing Analyst	Configure quotes, run scenarios, check margins
Sales Manager	Review and approve pricing requests
Finance Team	Track revenue leakage and cost accuracy
Business Executive	View overall pricing performance
Product Manager	Compare pricing scenarios for new products

Chapter 2 Business Problem

2.1 The Problem Today

Most small and mid-sized manufacturing businesses still calculate cost and price manually using spreadsheets. This creates several recurring problems.

- Manufacturing cost calculations are done manually and are error-prone.
- Selling prices are often based on guesswork rather than data.
- There is no easy way to compare multiple pricing scenarios before deciding.
- Discounts and manual overrides silently reduce profit margins.
- Business leaders do not have a single, real-time view of pricing performance.
- Price changes are not reviewed or approved in a structured way.
- There is no simple way to ask ”what if” questions about cost or price changes.

2.2 Impact of the Problem

Table 2.1: Business Impact of Manual Pricing

Problem Area	Business Impact
Manual cost calculation	Wasted time, calculation errors
Guesswork pricing	Lost revenue, uncompetitive pricing
No scenario comparison	Slow and risky decision-making
Uncontrolled discounts	Revenue leakage
No central dashboard	Poor visibility for leadership
No approval process	Inconsistent and unauthorized pricing

2.3 Why This Project Was Developed

This project was developed to solve the above problems using a single, easy-to-use platform. The goal is to replace manual spreadsheet work with an automated, data-driven, and AI-assisted pricing system.

Chapter 3 Technology Stack

3.1 Technologies Used

Table 3.1: Technology Stack of the Intelligent Pricing Platform

Technology	Category	Purpose in This Project
Snowflake	Cloud Data Warehouse	Stores cost, pricing, and transaction data
Streamlit	Web Application Framework	Builds the interactive user interface
Python	Programming Language	Core application logic and data processing
Gemini AI	AI / Language Model	Powers the AI Pricing Copilot and recommendations
Plotly	Data Visualization	Interactive charts for dashboards
Pandas	Data Processing Library	Cleans, transforms, and analyzes data

3.2 Why These Technologies Were Chosen

- **Snowflake** scales easily and separates storage from computing, which keeps the platform fast even as data grows.
- **Streamlit** allows a full web application to be built quickly using only Python, without needing a separate front-end framework.
- **Python** is used throughout the platform because it has strong libraries for data handling and AI integration.
- **Gemini AI** is used to understand natural language questions and generate pricing recommendations.
- **Plotly** provides interactive, business-friendly charts for dashboards.
- **Pandas** handles all in-memory data transformations before data is visualized or sent to the AI model.

Chapter 4 System Architecture

4.1 High-Level Architecture

The platform is organized into four layers that work together.

1. **Data Layer** – Snowflake stores all cost, product, pricing, and transaction data.
2. **Processing Layer** – Python and Pandas clean, transform, and calculate cost and pricing figures.
3. **Intelligence Layer** – Cortex AI generates pricing recommendations and answers user questions.
4. **Presentation Layer** – Streamlit and Plotly render the dashboards, forms, and charts that the user interacts with.

4.2 Component Responsibilities

Table 4.1: Component Responsibilities

Component	Responsibility
Streamlit UI	Screens, forms, navigation, user interaction
Python Business Logic	Cost calculation, pricing rules, workflow control
Snowflake	Data storage, historical records, reporting data
Cortex AI	Recommendations, chat-based Q&A, insight generation
Plotly	Interactive charts, trend lines, comparison graphs
Pandas	Data cleaning, aggregation, in-memory calculations

4.3 Design Principles

- Keep the user interface simple, even for non-technical users.
- Separate calculation logic from presentation logic.
- Keep all historical data in Snowflake for auditability.
- Use AI as a recommendation and support tool, not as the sole decision-maker.

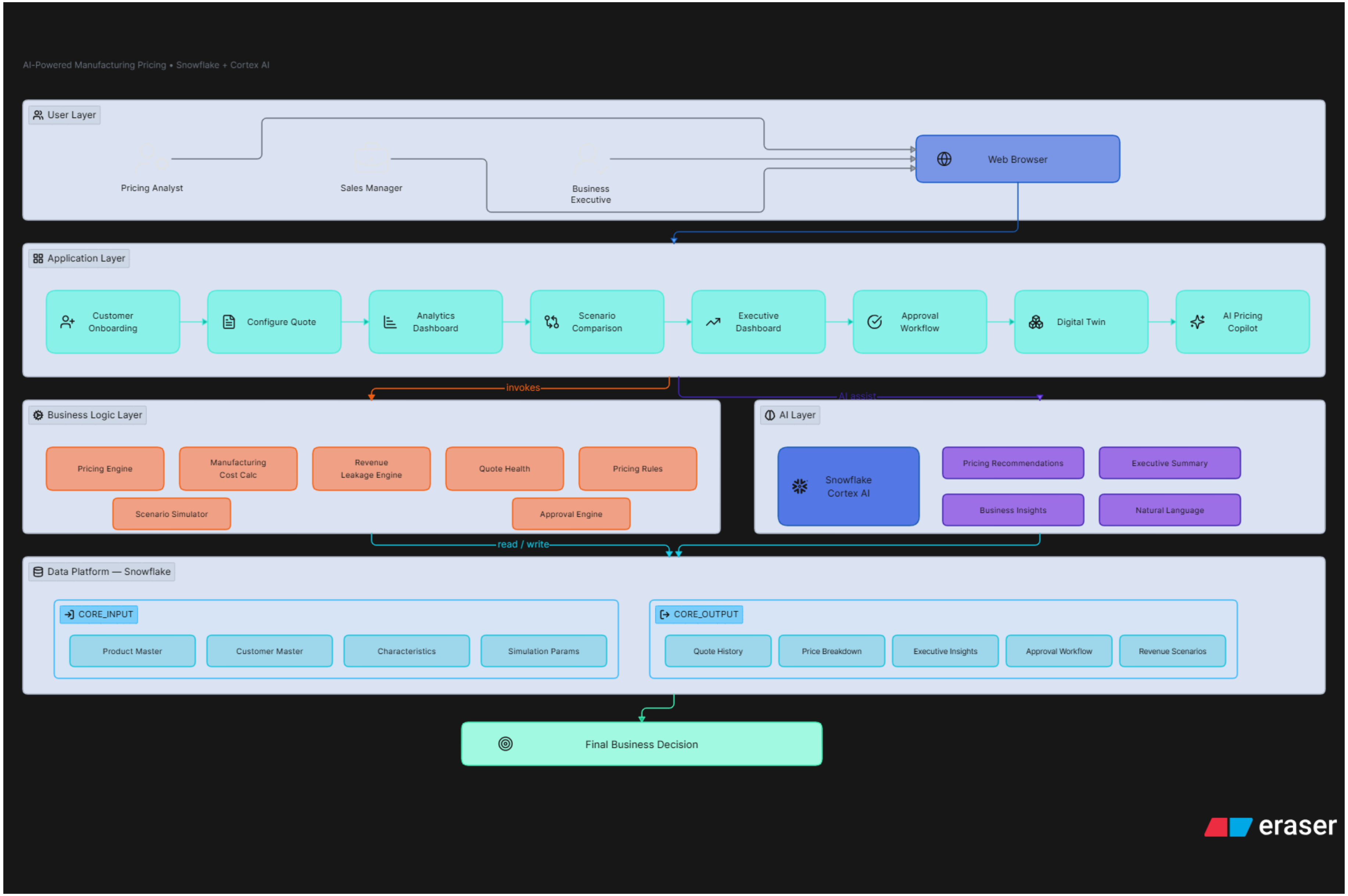

Figure 4.1: End-to-End Workflow of the Intelligent Pricing Platform

Chapter 5 Complete Workflow

5.1 End-to-End Customer Journey

1. Customer or product is onboarded into the system.
2. User configures a quote with cost inputs and target margin.
3. System calculates manufacturing cost and suggests a selling price.
4. User views the analytics dashboard for cost breakdown and trends.
5. User compares multiple pricing scenarios.
6. Business leadership reviews performance through the executive dashboard.
7. Pricing request goes through the approval workflow.
8. User can view a digital twin simulation of cost/price changes.
9. User can ask the AI Pricing Copilot for recommendations at any stage.

5.2 Workflow Diagram

Listing 5.1: End-to-End Workflow

```
Customer Onboarding
        |
  Configure Quote
        |
Analytics Dashboard <---> AI Pricing Copilot
        |
 Scenario Comparison
        |
 Approval Workflow
        |
 Executive Dashboard
        |
  Digital Twin View
```

5.3 Key Decision Points

- Should the price be approved, rejected, or sent back for revision?
- Which pricing scenario gives the best margin without losing competitiveness?
- Is there any revenue leakage that needs to be corrected?

Chapter 6 Customer Onboarding

6.1 Purpose

The Customer Onboarding module is the starting point of the Intelligent Pricing Platform. It allows the user to configure the application before generating any quotes. In this step, the user selects the pricing mode, connects to the required data source, and loads all the necessary input data into the system.

6.2 How it Works

The onboarding process consists of the following steps:

1. Select the pricing mode.
2. Connect the application to the required Snowflake database.
3. Choose the input method.
4. Load the required master and transaction data.
5. Validate the uploaded data.
6. Activate the platform and continue to the Configure Quote module.

6.3 Pricing Modes

The application supports two different pricing modes depending on the customer's requirements.

6.3.1 Integrated Pricing

Integrated Pricing is used when all pricing and manufacturing data is already available inside Snowflake. The application directly reads the required tables from the selected database and uses them for quote generation.

6.3.2 Standalone Pricing

Standalone Pricing is used when the customer does not have all the required data stored in Snowflake. In this mode, the user can upload the required datasets manually and perform pricing calculations without changing the application workflow.

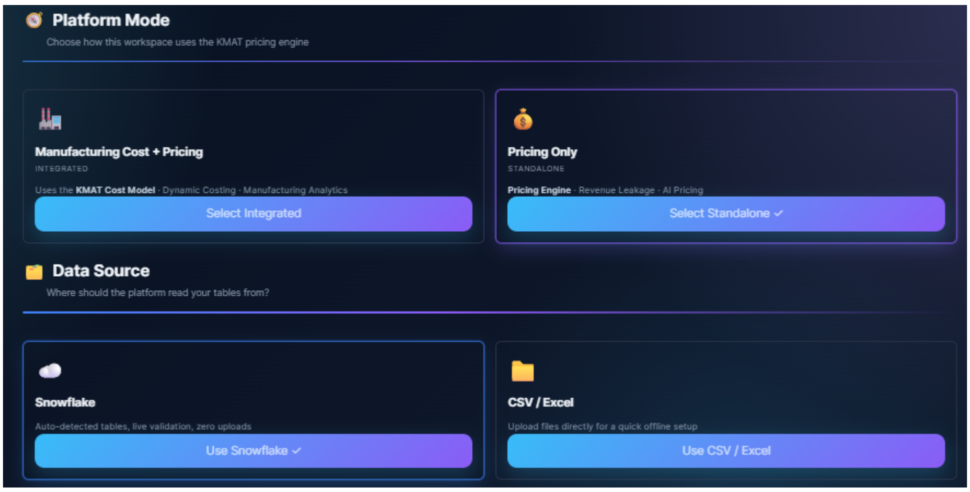

Figure 6.1: Selection of Integrated Pricing and Standalone Pricing modes.

6.4 Data Upload Options

The platform provides two methods for loading data.

- **Snowflake Tables** – The application directly connects to Snowflake and loads all required tables automatically.
- **CSV Files** – Users can upload CSV files containing the required master and pricing data. These files are validated before they are used by the application.

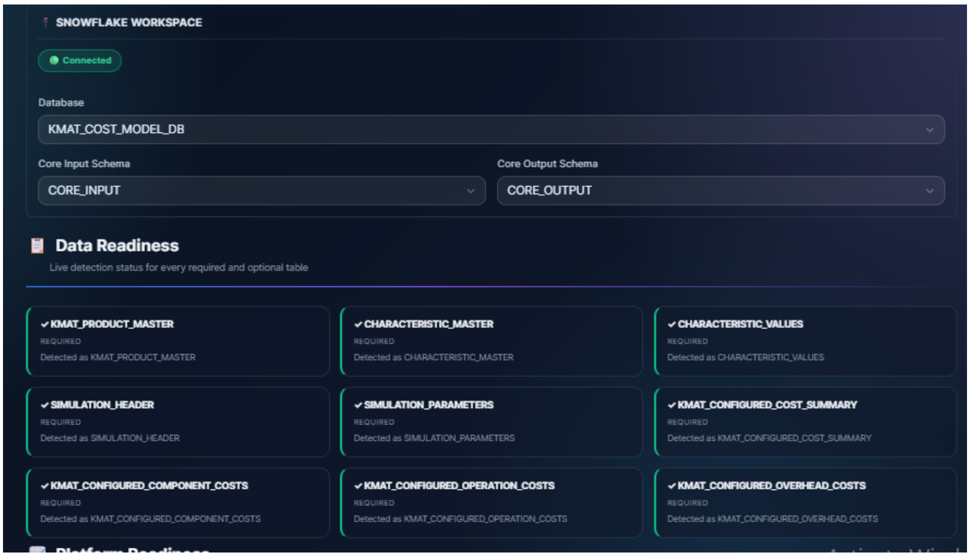

Figure 6.2: Snowflake workspace configuration and data loading.

6.5 Data Validation

Before activating the platform, the application checks that all required datasets are available and correctly loaded. The validation process includes:

- Checking the availability of required tables or CSV files.
- Verifying that mandatory columns are present.
- Displaying the loading status for each dataset.
- Reporting any missing or invalid data before proceeding.

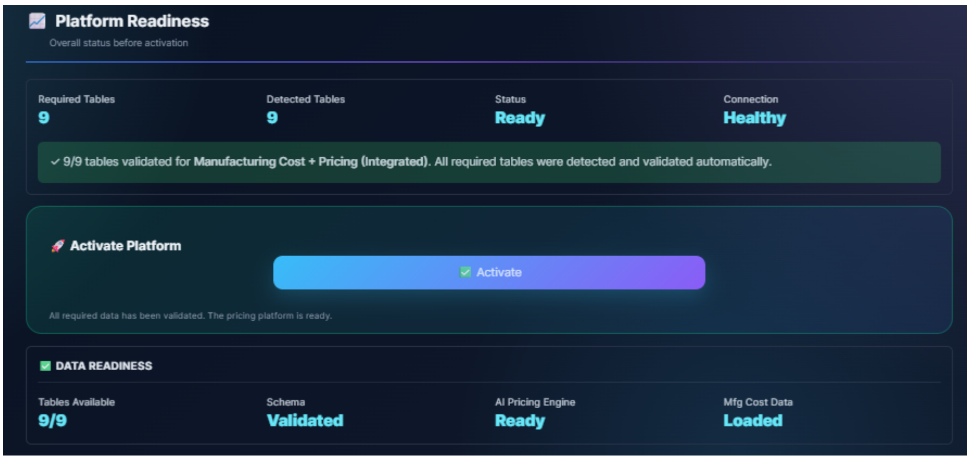

Figure 6.3: Platform readiness check and activation screen.

6.6 Workflow

1. Open the Customer Onboarding page.
2. Select either Integrated Pricing or Standalone Pricing.
3. Choose Snowflake Tables or CSV Upload as the data source.
4. Load all required datasets.
5. Review the data validation results.
6. Click **Activate Platform**.
7. Continue to the Configure Quote module.

6.7 Business Benefits

- Supports both integrated and standalone pricing workflows.
- Provides flexible data loading through Snowflake or CSV files.
- Ensures all required data is available before quote generation.
- Reduces manual errors through automatic validation.
- Creates a smooth and consistent starting point for the pricing process.

Chapter 7 Configure Quote

7.1 Purpose

The Configure Quote module is the core component of the Intelligent Pricing Platform. It allows users to configure a product, calculate the manufacturing cost, generate an optimized selling price, compare it with market prices, and evaluate the profitability of the quote before it is finalized.

This module combines cost calculation, pricing intelligence, business rules, and analytics into a single workflow.

7.2 Module Workflow

The quote generation process follows the steps below:

1. Select the customer and product.
2. Configure the product characteristics.
3. Enter the required order information.
4. Calculate the manufacturing cost.
5. Apply pricing rules and discounts.
6. Compare the quote with competitor prices.
7. Analyze profitability and revenue leakage.
8. Review AI pricing insights.
9. Save the generated quote.

7.3 Quote Configuration

This section collects all the information required to generate a quote.

The user selects:

- Customer
- Product
- Product characteristics
- Order quantity
- Region
- Currency

Once these values are selected, the application automatically retrieves the corresponding manufacturing data from Snowflake.

7.4 Configured Characteristics

Once a product is selected, the platform automatically displays the configured product characteristics. These characteristics are retrieved from the selected product configuration and are displayed in read-only mode to ensure consistency during quote generation.

Typical characteristics include:

- Engine Type
- Cab Type
- Product Color
- Wheel Configuration

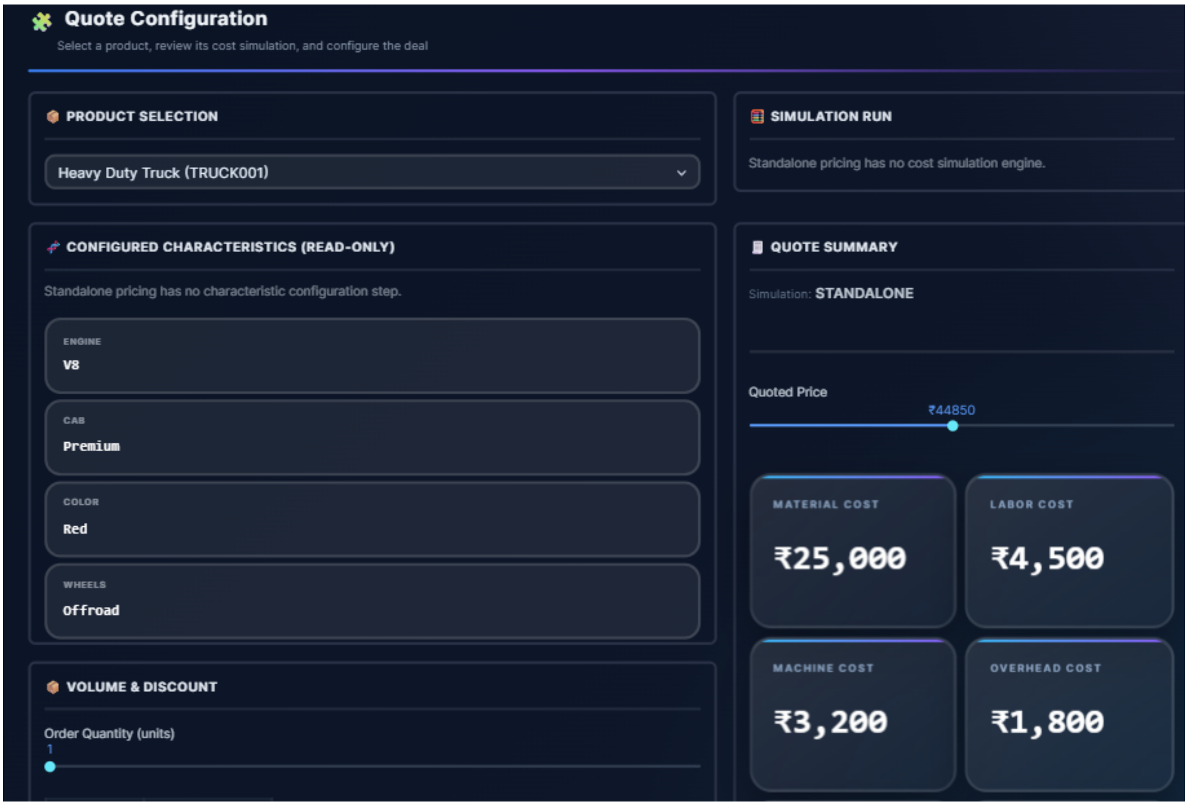

Figure 7.1: Quote Configuration Interface

The values shown in this section are used during pricing calculations and cannot be modified while generating the quote.

How to Interpret

- Displays the configuration selected for the current product.
- Ensures that pricing calculations are based on the correct product configuration.
- Prevents accidental modification of approved product specifications.

7.5 Pricing Summary

After calculating the manufacturing cost, the application generates the pricing summary.

This section displays:

- Manufacturing Cost
- Recommended Selling Price
- Target Margin
- Expected Profit
- Revenue Leakage

The recommended selling price is generated using the configured pricing logic and cost calculations.

Users can compare the recommended price with their own selling price before generating the final quote.

7.6 Volume and Discount Engine

The Volume and Discount Engine automatically applies discounts based on the order quantity. Larger order quantities qualify for higher discounts according to predefined pricing rules.

Users can also provide an additional manual discount when required.

The module displays:

- Order Quantity
- Applicable Volume Discount
- Discount Slab
- Additional Discount

How it Works

The system compares the entered order quantity with predefined discount slabs and automatically selects the applicable discount percentage.

How to Interpret

- Larger order quantities receive higher discounts.
- The selected discount tier is highlighted automatically.
- Additional discounts can be applied if approved by business policies.
- Every discount immediately updates the recommended selling price and expected profit.

The Discount Optimization Engine helps users apply discounts without reducing profitability.

Instead of applying random discounts, the application calculates how different discount percentages affect the final selling price and profit.

The engine provides:

- Recommended discount range
- Expected profit after discount
- Margin impact
- Revenue impact

This allows sales teams to negotiate confidently while maintaining healthy profit margins.

How to Interpret

- Higher discounts reduce the selling price.
- The application continuously recalculates profit.
- Users can immediately see whether the quote remains profitable.

7.7 Price Sensitivity Visualization

The Price Sensitivity Visualization shows how changes in selling price influence business performance.

It helps users understand how increasing or decreasing the selling price affects:

- Profit
- Margin
- Revenue Leakage
- Competitiveness

The visualization makes it easier to identify a balanced pricing strategy.

How to Interpret

- A higher selling price generally increases profit but may reduce competitiveness.
- A lower selling price may increase competitiveness but reduce profit.
- The ideal pricing point is where profitability and competitiveness are balanced.

7.8 Quote Health and Market Position

This section evaluates the quality of the generated quotation by combining profitability, revenue leakage, and market competitiveness.

The module displays:

- Revenue Leakage
- Quote Health Score
- Approval Status

Figure 7.2: Volume Discount Engine and Price Sensitivity Visualization

- Price Position Gauge
- Competitor Prices
- Market Recommendation

Revenue Leakage

Revenue Leakage indicates the difference between the recommended selling price and the quoted selling price. Lower leakage indicates better pricing performance.

Quote Health Score

The Quote Health Score measures the overall quality of the quotation.
It is calculated using multiple pricing factors including:

- Profit Margin
- Revenue Leakage
- Price Competitiveness
- Pricing Rules

Higher scores indicate healthier quotations.

Approval Status

Based on the health score and pricing rules, the system automatically determines whether the quote can be approved automatically or requires manual review.

Price Position Gauge

The Price Position Gauge compares the quoted price with competitor prices.
It indicates whether the quotation lies:

- Below Market
- Within Market Range
- Above Market

How to Interpret

- Healthy status indicates that the quote satisfies pricing rules.
- High health scores suggest low business risk.
- Quotes positioned close to the market average remain competitive.
- Low revenue leakage indicates better profitability.

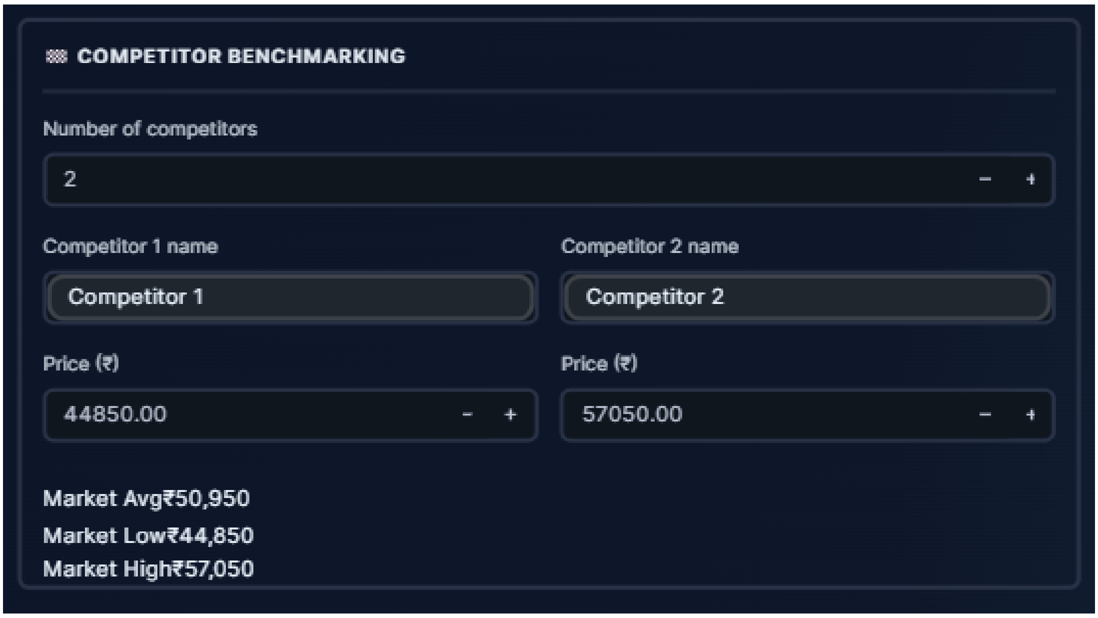

Figure 7.3: Competitor Benchmarking

7.9 Competitor Benchmarking

The platform allows users to compare the generated quote with competitor prices.
The comparison helps determine whether the current quote is:

- Above market price
- Within market range
- Below market price

This helps users make better pricing decisions before submitting a quotation.

7.10 Revenue Leakage Analysis

Revenue Leakage Analysis identifies situations where the selling price is lower than the expected profitable price.
The module highlights:

- Estimated revenue leakage
- Leakage percentage
- Margin risk
- Financial impact

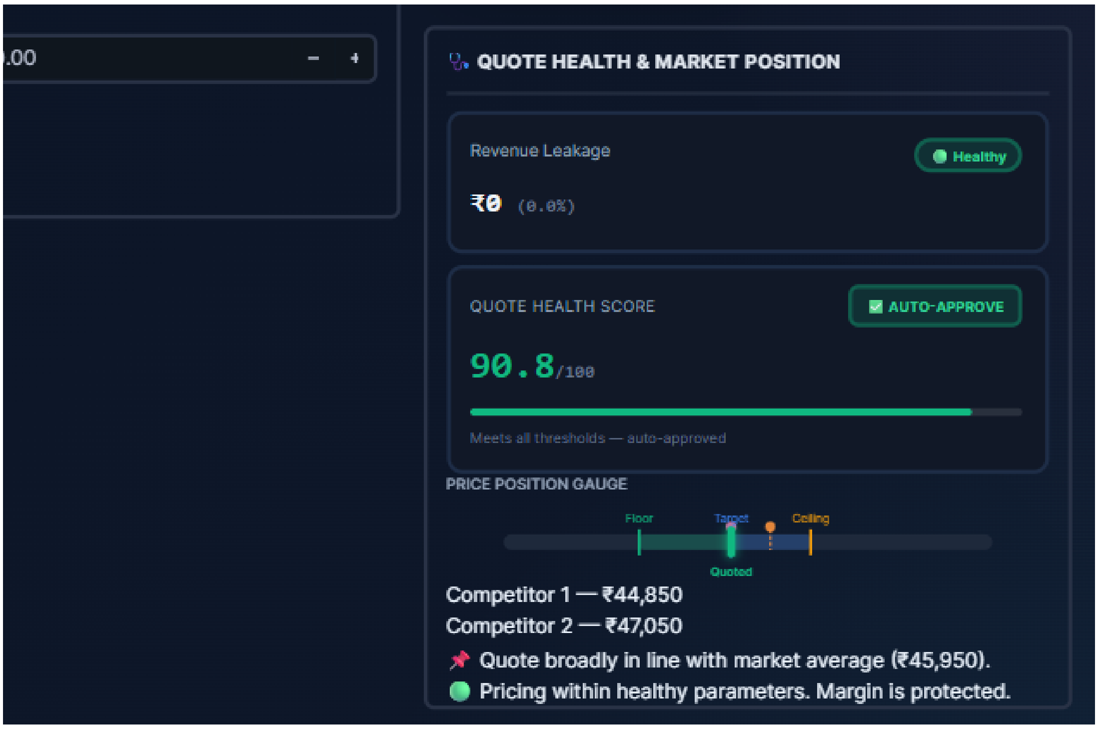

Figure 7.4: Quote Health, Revenue Leakage and Market Position

This enables pricing managers to identify quotes that may reduce business profitability.

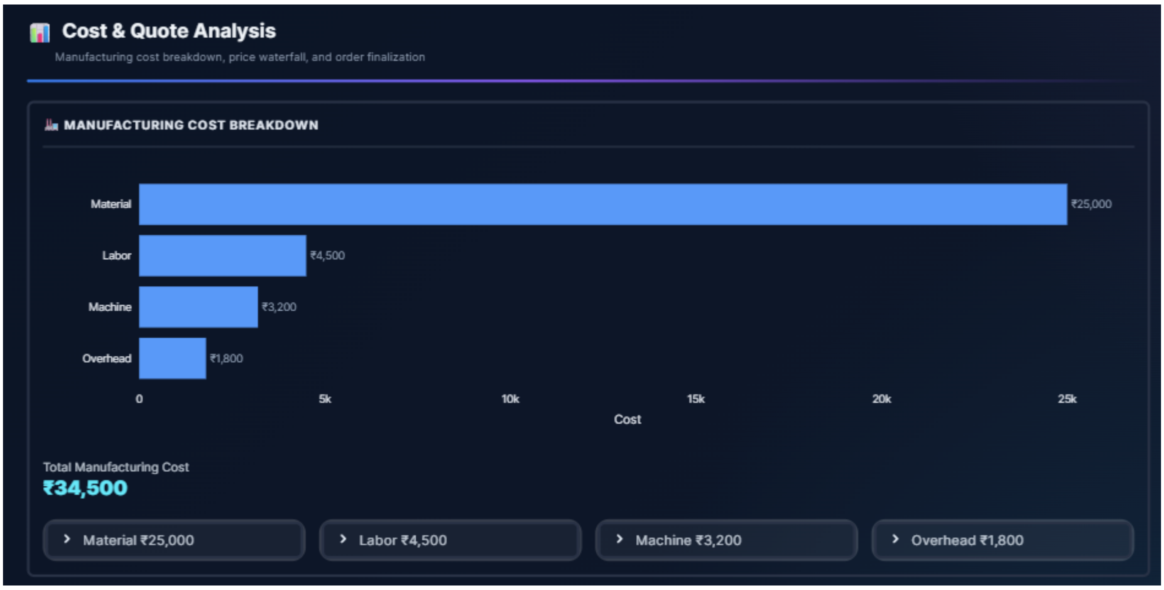

Figure 7.5: Pricing Summary and Cost Analysis

7.11 Business Benefits

- Automates the complete quotation process.
- Reduces manual pricing calculations.
- Improves pricing consistency.
- Optimizes discounts while protecting profitability.
- Helps identify revenue leakage before approving a quote.

Chapter 8 Analytics Dashboard

8.1 Purpose

The Analytics Dashboard provides a complete overview of the organization's pricing performance. It combines key business metrics, revenue analysis, pricing insights, risk indicators, and AI-generated recommendations into a single dashboard. This helps business users monitor quote performance, identify revenue leakage, and make informed pricing decisions.

8.2 How it Works

After a quote is generated, the application stores all pricing and cost information in Snowflake. The Analytics Dashboard retrieves this data, processes it using Python and Pandas, and displays the results through interactive charts and KPI cards.

Users can filter the dashboard based on:

- Product
- Customer
- Risk Level
- Approval Status
- Date Range

The dashboard updates automatically whenever the selected filters are changed.

8.3 Performance Overview

The Performance Overview section provides a quick summary of the overall pricing portfolio.

It displays important Key Performance Indicators (KPIs), including:

- Total Quotes Generated
- Average Target Price
- Average Actual Price
- Total Revenue Leakage
- Average Achieved Margin
- High-Risk Quotes
- Warning Quotes
- Healthy Quotes

How to Interpret

- A higher average margin indicates better profitability.
- Revenue leakage shows the total amount of potential revenue lost due to lower selling prices.
- High-risk and warning quotes require further review before approval.
- Healthy quotes meet the expected pricing objectives.

8.4 Executive Pricing Insights

This section provides a summarized business view of the current pricing performance.

The dashboard automatically highlights:

- Total Quotes Generated
- Revenue Leakage Detected

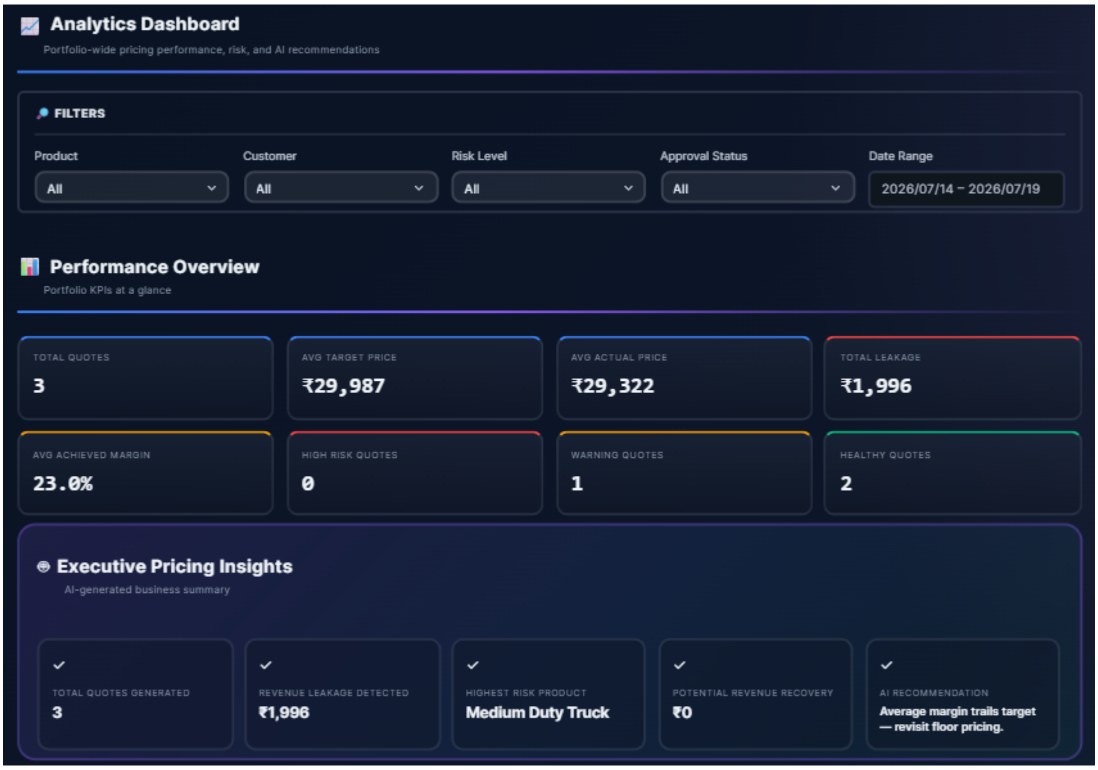

Figure 8.1: Performance Overview and Dashboard Filters

- Highest Risk Product
- Potential Revenue Recovery
- AI-generated Pricing Recommendation

Instead of manually reviewing every quote, managers can quickly understand the overall business situation from this summary.

How to Interpret

The AI recommendation provides a short business suggestion based on the current pricing performance. It helps decision makers identify products or quotes that may require immediate attention.

8.5 Business Performance

This section compares pricing performance across different products.

It contains two visualizations:

Quote Performance

This chart compares the Target Price with the Actual Selling Price for every product.

It helps users identify products that are being sold below the recommended price.

Cost Distribution

The Cost Distribution chart shows the contribution of each manufacturing cost component, including:

- Material Cost
- Machine Cost
- Setup Cost
- Overhead Cost

How to Interpret

- A larger Material Cost indicates that raw materials contribute the highest share of manufacturing cost.
- Comparing target and actual prices helps identify products that may cause revenue leakage.

Figure 8.2: Business Performance Charts

8.6 Risk Analytics

The Risk Analytics section identifies quotes that may reduce profitability.

The Leakage Risk Register displays:

- Product Name
- Order Quantity
- Target Price
- Actual Price
- Revenue Leakage
- Leakage Percentage
- Risk Level
- Health Score
- Approval Routing

Users can also filter the table based on leakage percentage and minimum selling price.

The Leakage by Product chart highlights products contributing the highest revenue leakage.

How to Interpret

- Higher leakage percentage indicates greater revenue loss.
- Low health scores indicate quotes requiring manager review.
- Products appearing at the top of the leakage chart should be prioritized for pricing correction.

8.7 Business Activity

The Business Activity section provides an overview of recent quotations and product performance.

It includes:

- Recent Quotes
- Approval Status
- Margin Achieved
- Top Performing Products
- Revenue Generated
- Average Margin
- Quote Count

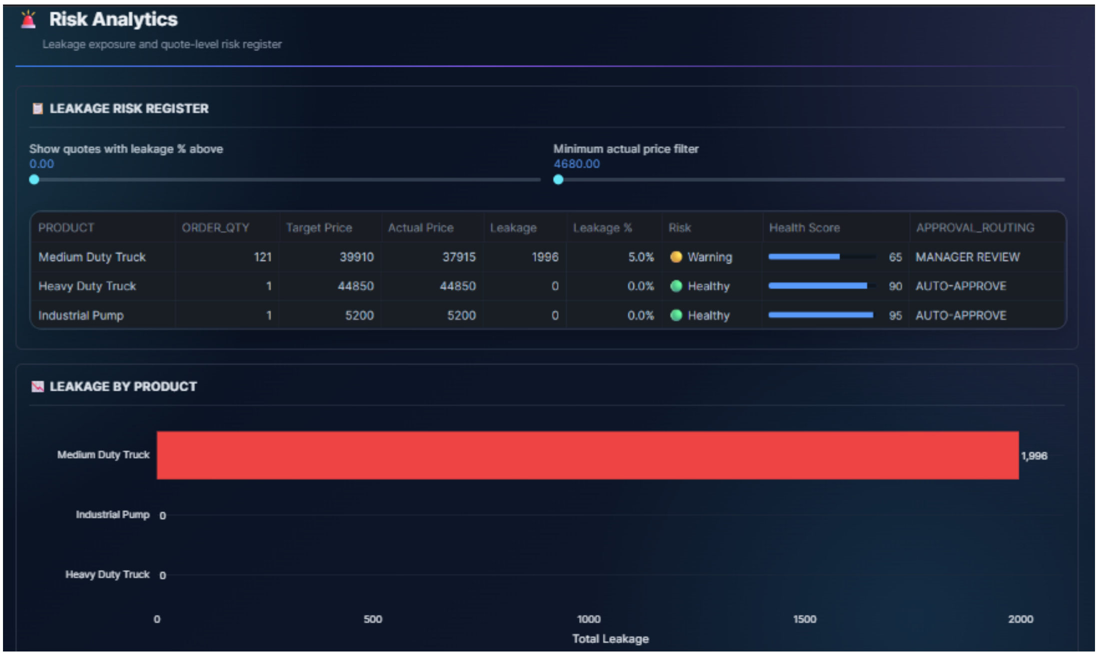

Figure 8.3: Risk Analytics Dashboard

- Product Health Percentage

How to Interpret

This section helps business users understand which products are performing well and which products require additional attention.

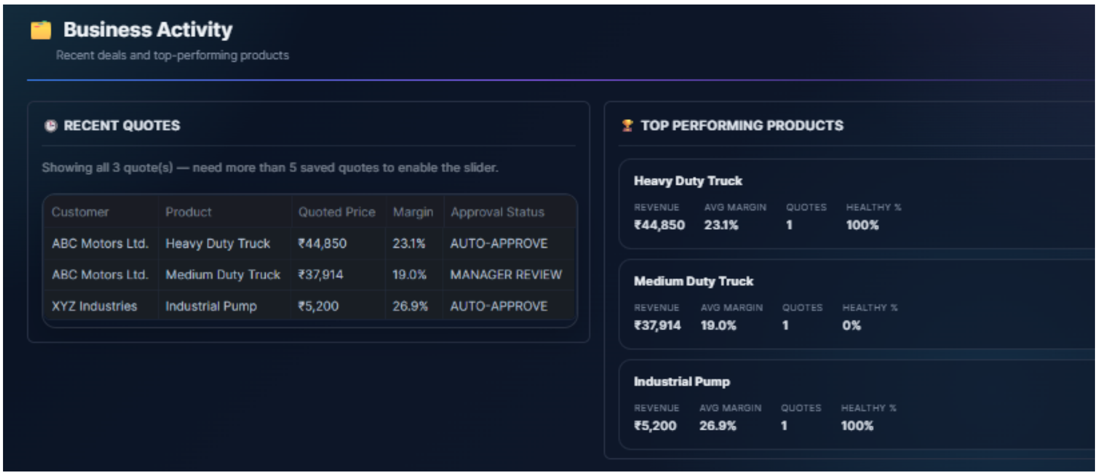

Figure 8.4: Business Activity

8.8 Business Benefits

- Provides a complete overview of pricing performance.
- Helps identify revenue leakage early.
- Supports better pricing decisions through visual analytics.
- Enables faster executive decision making using AI-generated insights.
- Tracks product performance and approval status in one place.
- Simplifies monitoring through interactive filters and dashboards.

Chapter 9 Scenario Comparison

9.1 Purpose

The Scenario Comparison module allows users to compare multiple pricing scenarios side by side before choosing one.

9.2 How it Works

- The user creates two or more pricing scenarios with different cost or margin assumptions.
- The system calculates the resulting price and profit for each scenario.
- Results are displayed side by side for comparison.

9.3 Key Features

- Side-by-side scenario comparison table.
- Visual comparison charts.
- Ability to save and reload scenarios.
- Highlighting of the most profitable option.

9.4 Workflow

1. Create Scenario A and Scenario B (or more).
2. Adjust cost or margin assumptions for each.
3. Compare calculated prices and margins.
4. Select the preferred scenario.

9.5 Business Benefits

- Reduces risk of choosing a weak pricing option.
- Speeds up decision-making for sales teams.
- Encourages data-driven negotiation with customers.

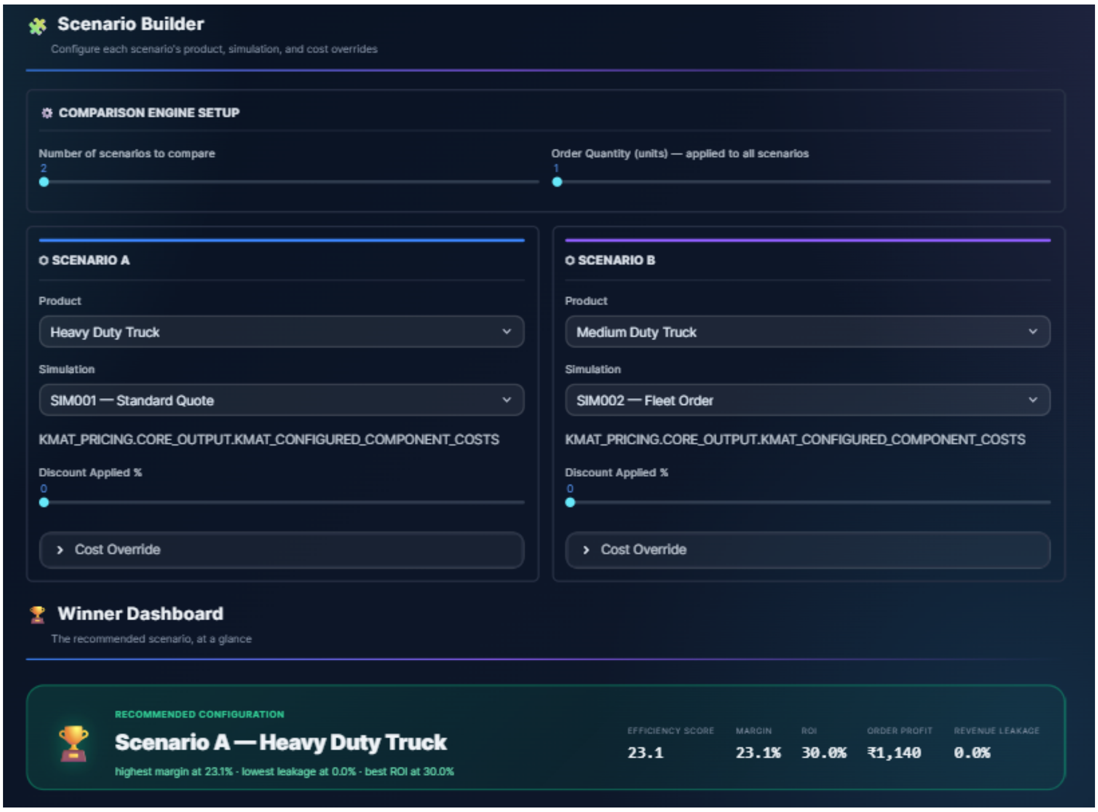

Figure 9.1: Scenario Comparison

Figure 9.2: Visual Analytics for comparison

Chapter 10 Executive Dashboard

10.1 Purpose

The Executive Dashboard provides a high-level summary of the overall pricing performance of the organization. It combines important business metrics, revenue analysis, risk indicators, and performance trends into a single view. This dashboard helps executives quickly understand the health of the pricing process and identify areas that require attention.

10.2 How it Works

The dashboard collects data from all generated quotes stored in Snowflake and calculates important business metrics. These metrics are updated automatically whenever new quotes are generated or existing quotes are modified.

The dashboard consists of three main sections:

- Organization-Wide Intelligence
- Performance Analytics
- Product Performance Summary

10.3 Organization-Wide Intelligence

The first section provides an overview of the organization's pricing performance using Key Performance Indicators (KPIs).

Total Quotes

Displays the total number of quotations generated during the selected period.

Calculation

$$\text{Total Quotes} = \text{Number of Generated Quotes}$$

Interpretation

A higher value indicates increased business activity and customer engagement.

Total Quoted Revenue

Displays the total revenue generated from all quotations.

Calculation

$$\text{Total Revenue} = \sum \text{Quoted Price}$$

Interpretation

This metric represents the overall business value generated by the quotations.

Total Revenue Leakage

Shows the total revenue lost because the quoted price is lower than the recommended target price.

Calculation

$$\text{Revenue Leakage} = \sum (\text{Target Price} - \text{Actual Price})$$

(Only when Actual Price is lower than Target Price.)
Interpretation
Lower revenue leakage indicates better pricing decisions.

Average Achieved Margin

Displays the average profit margin achieved across all quotations.
Calculation

$$\frac{\sum Margin\%}{Number\ of\ Quotes}$$

Interpretation
This value should remain close to or above the target margin.

Healthy Quotes

Shows the number of quotations that satisfy pricing and profitability requirements.
Higher values indicate better overall pricing quality.

High-Risk Quotes

Displays quotations having high revenue leakage or poor profit margin.
These quotes usually require additional review before approval.

Pending Approvals

Shows the number of quotations waiting for management approval.

Average Quote Health Score

Represents the overall quality of generated quotations.
The health score considers:
- Profit Margin
- Revenue Leakage
- Risk Level
- Pricing Compliance

Higher scores indicate healthier quotations.

10.4 Performance Analytics

This section provides visual insights into pricing performance.

Revenue vs Leakage Trend

This graph compares total quoted revenue with revenue leakage over time.
Interpretation
- Increasing revenue indicates business growth.
- Increasing leakage indicates pricing inefficiencies.

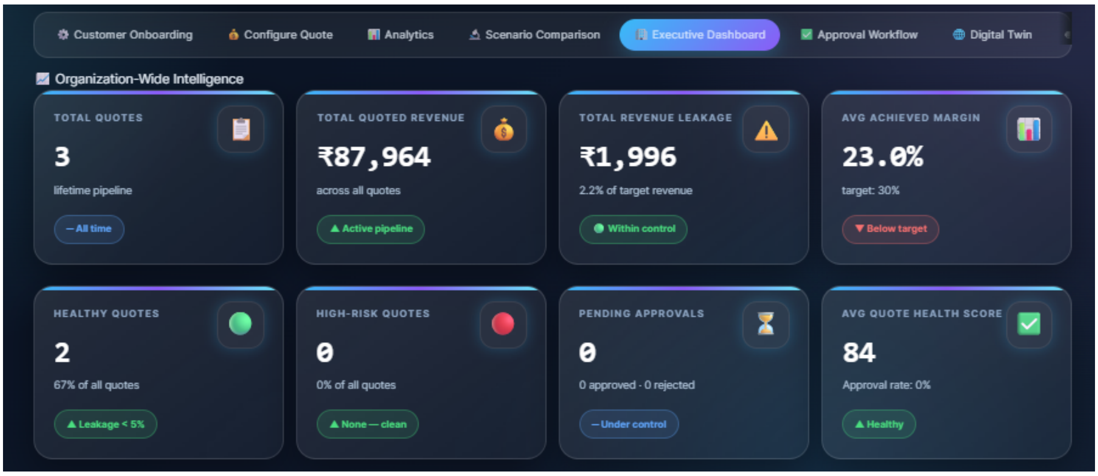

Figure 10.1: Organization-Wide Intelligence Dashboard

- The objective is to increase revenue while keeping leakage as low as possible.

Risk Distribution

This chart displays the percentage of:

- Healthy Quotes
- Warning Quotes
- High-Risk Quotes

Interpretation

A larger Healthy section indicates a stable pricing portfolio, while an increase in Warning or High-Risk quotes suggests that pricing policies should be reviewed.

Quote Health Score Distribution

This chart shows how quote health scores are distributed.

Interpretation

Most quotes should have high health scores. Lower scores indicate quotations requiring further analysis.

Approval Routing

Displays how quotations are routed through the approval workflow.

Possible routing includes:

- Auto Approve
- Manager Review
- Director Review

Interpretation

A higher number of Manager or Director reviews indicates that more quotations require manual verification.

10.5 Product Performance Summary

This table summarizes the pricing performance of individual products.

It displays:

- Product Name

Figure 10.2: Performance Analytics

- Number of Quotes
- Total Revenue
- Total Revenue Leakage
- Average Selling Price
- Average Health Score
- Leakage Percentage

Interpretation

The table helps identify products that generate the highest revenue and products responsible for the highest revenue leakage.

Product Performance Summary

PRODUCT	Quotes	Total_Revenue	Total_Leakage	Avg_Price	Avg_Health	Leakage Rate %
Heavy Duty Truck	1	₹44,850	₹0	₹44,850	90.8	0
Industrial Pump	1	₹5,200	₹0	₹5,200	95.9	0
Medium Duty Truck	1	₹37,914	₹1,996	₹37,914	65.4	5.3

Figure 10.3: Product Performance Summary

10.6 Margin Achievement Trend

The Margin Achievement Trend compares the achieved profit margin with the target margin over time.

Interpretation

- When the achieved margin remains close to the target margin, pricing performance is considered healthy.
- A continuous drop below the target margin indicates that pricing strategies may require adjustment.

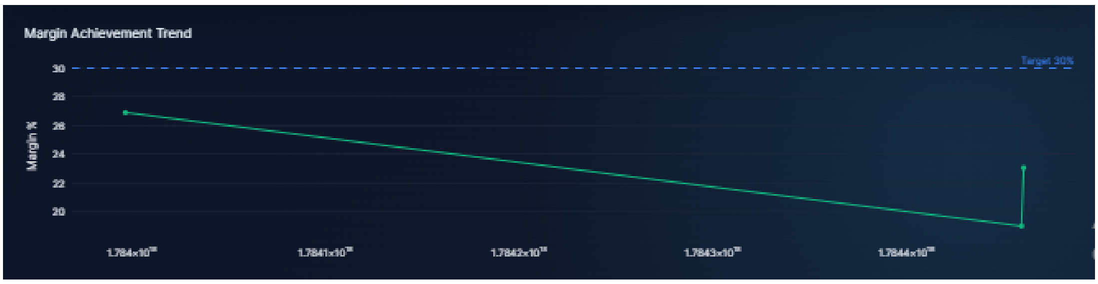

Figure 10.4: Margin Achievement Trend

10.7 Business Benefits

- Provides executives with a complete view of organizational pricing performance.
- Helps monitor revenue, profitability, and pricing health.
- Identifies revenue leakage before it impacts business performance.
- Supports faster decision-making through interactive visualizations.
- Enables management to prioritize high-risk quotations.
- Tracks product performance and approval status in a single dashboard.

Chapter 11 Digital Twin

11.1 Purpose

The Digital Twin module creates a simulated model of the pricing and cost structure, allowing users to test changes safely before applying them.

11.2 How it Works

- The system builds a virtual model using current cost and pricing data.
- The user changes an input (e.g., raw material cost increase).
- The system simulates the impact on price and margin without affecting live data.

11.3 Key Features

- Real-time ”what if” simulation.
- Visual representation of impact on cost and margin.
- Safe environment that does not affect live pricing data.
- Side-by-side view of current vs. simulated state.

11.4 Workflow

1. Open the Digital Twin view.
2. Adjust a cost or pricing input.
3. View the simulated impact instantly.
4. Decide whether to apply the change in the live system.

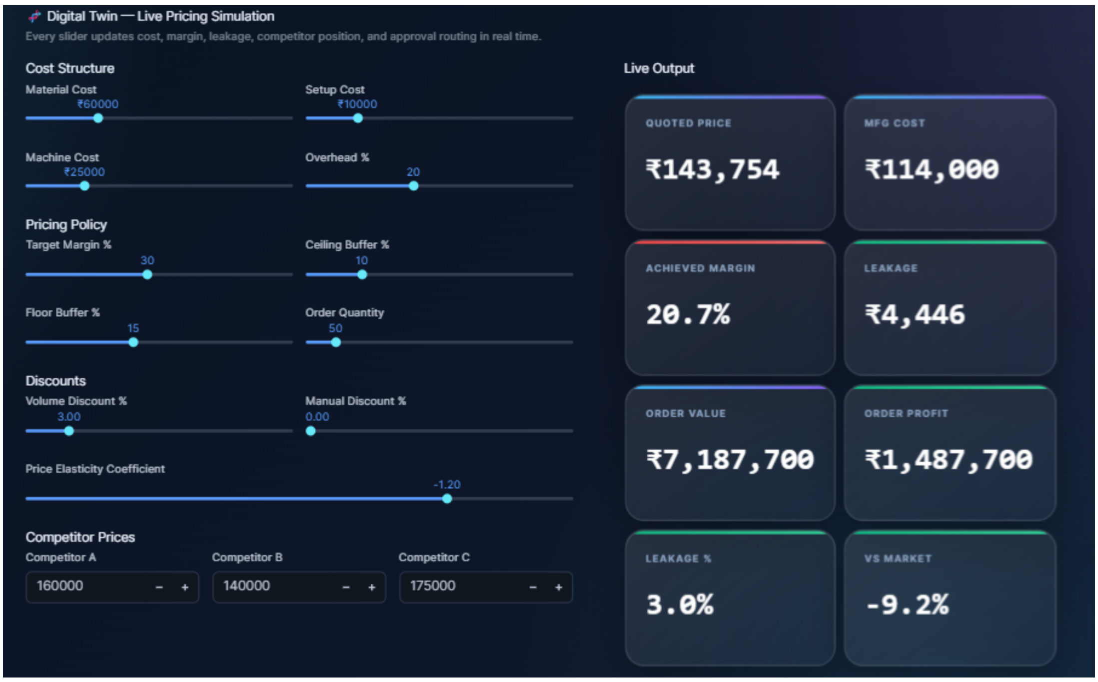

Figure 11.1: Digital Twin

11.5 Price Position Analysis

The Price Position Gauge compares the quoted price with competitor prices available in the market.

The gauge displays:

- Floor Price
- Target Price
- Ceiling Price
- Competitor Prices
- Current Quoted Price

Figure 11.2: Price Position Analysis

11.6 Elasticity Simulation

The Elasticity Simulation predicts how changes in selling price affect customer demand and business revenue.

The simulation displays:

- Selling Price
- Expected Demand
- Expected Revenue
- Revenue Change
- Expected Margin

How to Interpret

- Lower prices generally increase customer demand.
- Higher prices improve profit margin but may reduce demand.
- The objective is to identify a price that maximizes both revenue and profitability.

11.7 Margin Heatmap

The Margin Heatmap visualizes the relationship between discount percentage and order quantity.

Each cell represents the expected profit margin for a particular combination of:

- Discount Percentage
- Order Quantity

The color scale helps users quickly identify profitable and less profitable pricing scenarios.

How to Interpret

- Green regions indicate healthy profit margins.
- Yellow regions indicate acceptable margins.

- Red regions indicate pricing scenarios that should generally be avoided.

Elasticity — Demand & Revenue at Price Variants

Scenario	Price	Demand (units)	Revenue	Rev Δ	Margin %
-20%	₹115,003	124	₹14,260,397	-0.8%	0.9%
-10%	₹129,379	112	₹14,490,403	+0.8%	11.9%
-5%	₹136,566	106	₹14,476,028	+0.7%	16.5%
Current	₹143,754	100	₹14,375,400	+0.0%	20.7%
+5%	₹150,942	94	₹14,188,520	-1.3%	24.5%
+10%	₹158,129	88	₹13,915,387	-3.2%	27.9%
+20%	₹172,505	76	₹13,110,365	-8.8%	33.9%

Figure 11.3: Elasticity Simulation and Margin Heatmap

11.8 Business Benefits

- Reduces risk when testing cost or pricing changes.
- Helps anticipate the impact of market or supplier changes.
- Improves confidence before making live pricing decisions.

Chapter 12 AI Pricing Copilot

12.1 Purpose

The AI Pricing Copilot allows users to ask pricing-related questions in plain language and receive AI-generated insights and recommendations.

12.2 How it Works

- The user types a question or request in natural language.
- The request, along with relevant data, is sent to Cortex AI.
- Cortex AI generates a response or recommendation.
- The response is displayed to the user inside the platform.

12.3 Key Features

- Natural language question and answer interface.
- AI-generated pricing recommendations.
- Explanation of cost or margin trends in simple language.
- Quick suggestions during quote configuration.

12.4 Workflow

1. Open the AI Pricing Copilot chat panel.
2. Ask a question (e.g., ”Why did margin drop this month?”).
3. Cortex AI analyzes relevant data.
4. The copilot responds with an explanation or recommendation.

12.5 Business Benefits

- Makes pricing insights accessible to non-technical users.
- Speeds up decision-making with instant answers.
- Reduces dependency on manual analysis for common questions.

Chapter 13 Database Overview

13.1 Database Architecture

The Intelligent Pricing Platform uses Snowflake as its primary cloud database. The database is organized into two main schemas:

- **CORE_INPUT** – Stores all master data, simulation inputs, and cost calculation views.
- **CORE_OUTPUT** – Stores generated quotations, pricing results, approval workflows, executive insights, and scenario outputs.

This separation keeps the input data independent from the generated business results and improves maintainability of the application.

13.2 CORE_INPUT Schema

The CORE_INPUT schema contains all master data required for generating quotations and performing pricing calculations.

Input Tables

Table 13.1: CORE_INPUT Tables

Table	Purpose
CHARACTERISTIC_MASTER	Stores the list of configurable product characteristics.
CHARACTERISTIC_VALUES	Stores possible values for each product characteristic.
CUSTOMER_MASTER	Stores customer information used during quote generation.
KMAT_PRODUCT_MASTER	Stores configurable KMAT product details.
SIMULATION_HEADER	Stores general information for each pricing simulation.
SIMULATION_PARAMETERS	Stores user-selected simulation parameters used during pricing calculations.

Input Views

The application uses Snowflake views to simplify pricing calculations and avoid repeated data processing.

Table 13.2: CORE_INPUT Views

View	Purpose
VW_COMPONENT_COSTS	Displays material and component-level manufacturing costs.
VW_COST_SUMMARY	Provides consolidated manufacturing cost for configured products.
VW_OPERATION_COSTS	Stores operation and processing costs.
VW_OVERHEAD_COSTS	Calculates manufacturing overhead expenses.
VW_PRICING_INPUT	Combines all required pricing inputs into a single view.
VW_SIMULATION_CONFIGURATION	Displays the complete product configuration selected for simulation.

13.3 CORE_OUTPUT Schema

The CORE_OUTPUT schema stores all generated outputs after pricing calculations are completed.

Table 13.3: CORE_OUTPUT Tables

Table	Purpose
APPROVAL_WORKFLOW	Stores approval status and routing information for generated quotations.
EXECUTIVE_INSIGHTS	Stores executive-level business insights generated by the platform.
PRICE_BREAKDOWN	Stores detailed manufacturing cost and pricing breakdown for each quote.
PRICE_SIMULATION_SUMMARY	Stores summary results of pricing simulations.
QUOTE_HISTORY	Stores previously generated quotations for reporting and analytics.
REVENUE_SCENARIOS	Stores multiple pricing scenarios used for comparison and optimization.

13.4 Database Workflow

The following workflow explains how data moves through the application.

1. Product, customer, and configuration data are loaded into the CORE_INPUT schema.
2. Snowflake views calculate component costs, operation costs, overhead costs, and pricing inputs.
3. The Configure Quote module reads these views to generate manufacturing costs and recommended prices.
4. Generated quotations and pricing results are stored in the CORE_OUTPUT schema.
5. Analytics Dashboard, Executive Dashboard, Approval Workflow, Digital Twin, and AI Pricing Copilot read the output tables to display business insights.

Chapter 14 Current Project Status

14.1 Completed Features

Table 14.1: Completed Features

Feature	Status
Customer Onboarding	Completed
Configure Quote	Completed
Analytics Dashboard	Completed
Manufacturing Cost Calculation	Completed
Executive Dashboard	Completed
Scenario Comparison	Completed
Digital Twin	Completed

14.2 Ongoing Work

Table 14.2: Ongoing Features

Feature	Status
AI Pricing Copilot	In Progress

14.3 Future Enhancements

Table 14.3: Planned Future Enhancements

Feature	Planned Status
Mobile-Friendly View	Planned

Chapter 15 Challenges Faced

15.1 Key Challenges

Table 15.1: Challenges Faced During Development

Challenge	How It Was Addressed
Inconsistent cost data formats	Standardized data cleaning with Pandas
Designing a simple UI for non-technical users	Iterative Streamlit UI design
Structuring AI prompts for reliable output	Refined prompt design for Gemini AI
Handling multiple pricing scenarios efficiently	Optimized scenario data model
Ensuring smooth Snowflake integration	Careful schema and query design

Chapter 16 Future Enhancements

16.1 Planned Improvements

- **Advanced AI Pricing Copilot**: Add support for deeper, multi-step pricing analysis and proactive alerts.
- **Full Executive Dashboard**: Expand KPIs to include forecasting and year-over-year comparisons.
- **Mobile-Friendly View**: Optimize the interface for tablets and mobile devices.
- **Role-Based Access Control**: Add fine-grained permissions for different user roles.
- **Automated Revenue Leakage Alerts**: Notify users automatically when leakage is detected.
- **Integration with ERP Systems**: Allow direct data exchange with external ERP platforms.

16.2 Long-Term Vision

The long-term vision is to evolve the platform into a complete, AI-assisted pricing command center that can be used across multiple business units, with predictive pricing recommendations and deeper integration into existing enterprise systems.

Chapter 17 Conclusion

17.1 Summary

The Intelligent Pricing Platform brings structure, automation, and AI-driven intelligence to a process that is traditionally manual and error-prone. By combining Snowflake, Streamlit, Python, Gemini AI, Plotly, and Pandas, the platform provides:

- Accurate and automated manufacturing cost calculation.
- Data-driven, optimized pricing recommendations.
- Clear scenario comparison for better decisions.
- Visibility into revenue leakage.
- A structured approval process for pricing changes.
- A safe simulation environment through the digital twin.
- An AI copilot that makes pricing insights accessible to everyone.

17.2 Closing Note

This document reflects the current design and status of the Intelligent Pricing Platform as of the time of writing. It will continue to evolve as new features are completed and new requirements are identified.